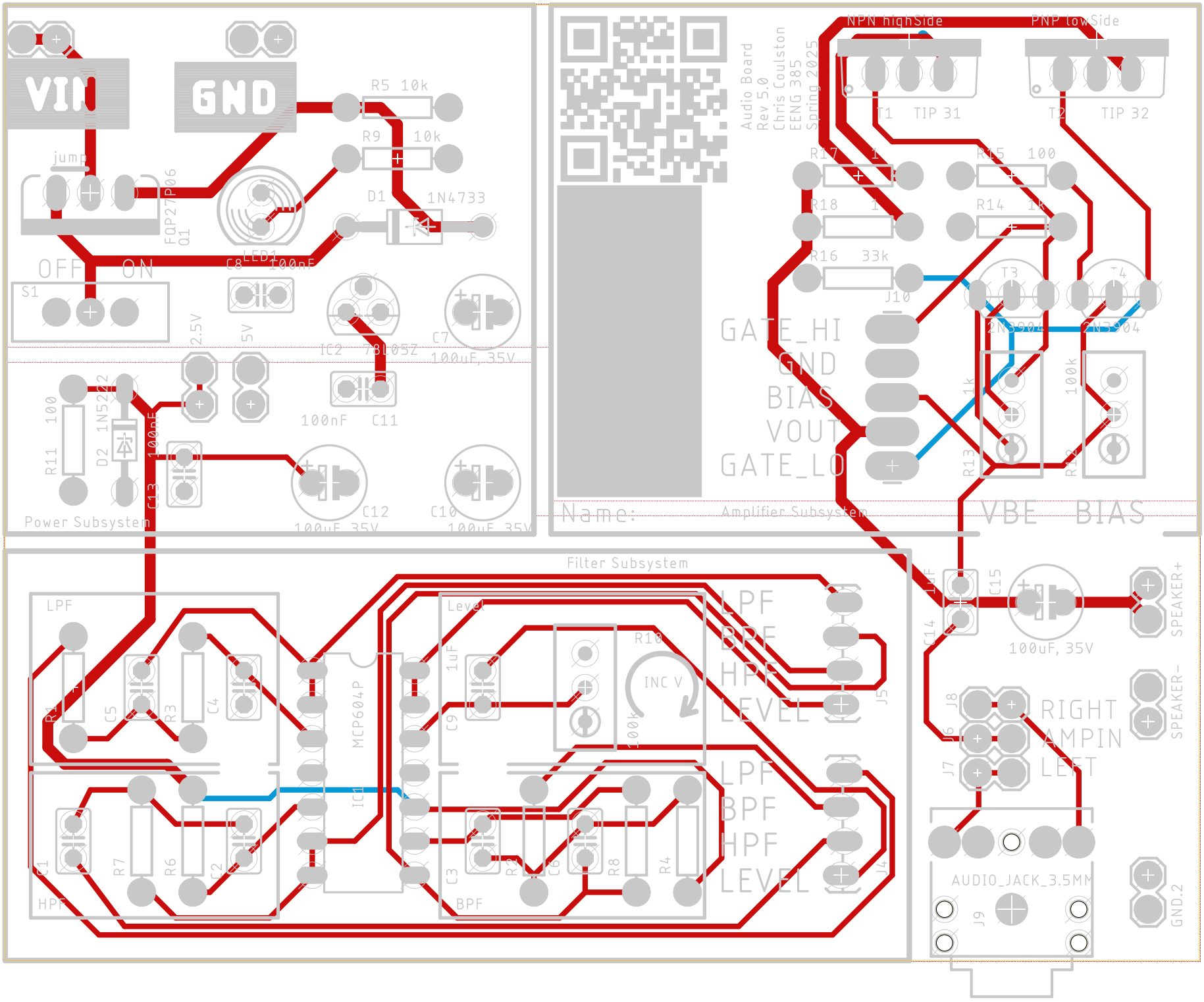

VIN
GND
jump
R5 10k
R9 10k
D1
1N4733
FQP27P06
Q1
LED1
C8 100nF
OFF
ON
S1
2.5V
5V
IC2 78L05Z
C7
100uF 35V
100nF C11
R11 100
D2 1N5242
C13 100nF
C12
100uF 35V
C10
100uF 35V
Power Subsystem
Audio Board
Rev 5.0
Chris Coulston
EENG 385
Spring 2025
NPN highSide
PNP lowSide
T1 TIP 31
T2 TIP 32
R17 1
R18 1
R15 100
R14 1k
R16 33k
T3
T4
2N3904
2N3904
J10
GATE_HI
GND
BIAS
VOUT
GATE_LO
R13 1k
100k
Name:
Amplifier Subsystem
VBE BIAS
Filter Subsystem
LPF
HPF
MCP604P
IC1
Level
1uF
C9
R10
INC V
100k
J5
LPF
BPF
HPF
LEVEL
C1
R7
R6
C2
C3
R2
C6
R8
R4
J4
C14 1uF
C15
100uF, 35V
SPEAKER+
SPEAKER-
J8
J6
J7
RIGHT
AMPIN
LEFT
AUDIO_JACK_3.5MM
J9
GND.2